Feature engineer

Interview Explanation for Each Feature

1. Year

Extracted from the Date column to capture yearly trends and long-term changes in stock prices.

2. Month

Extracted to identify seasonal effects and monthly stock market patterns.

3. Day

Represents the day of the month and helps detect recurring daily patterns.

4. Day_of_Week

Indicates the trading weekday, which may influence stock behavior due to market activity.

5. Previous_Close

Stores the previous trading day's closing price, providing historical context for prediction.

6. Close_Lag_3

Stores the closing price from three trading days earlier to capture recent historical trends.

7. Rolling_Mean_7

Calculates the average closing price over the previous seven days, reducing noise and showing the overall trend.

8. Rolling_STD_7

Calculates the standard deviation over the previous seven days to measure stock price volatility and risk.

9. Closing_Price_Difference

Calculates the difference between today's and yesterday's closing price, showing the daily price movement.

10. Daily_Return

Calculates the percentage change in closing price from the previous day, making it easier to compare performance across companies.

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Project Title

Stock Market Price Prediction Using Machine Learning

Project Overview

This project predicts future stock price movements using historical stock market data collected from **Yahoo Finance**. The data is cleaned, analyzed, and enhanced through feature engineering before training multiple machine learning models. The models predict the future **closing price**, which is used to determine whether the stock price is likely to move **upward or downward**. Among the tested models, **XGBoost** achieved the best performance.

Use Cases

This project has several practical applications in the financial domain:

1. **Stock Price Trend Prediction:** Predicts whether the stock price is likely to move **upward or downward** based on historical market data.
2. **Investment Decision Support:** Helps investors make better buy, hold, or sell decisions by estimating future stock price movements.
3. **Market Trend Analysis:** Assists financial analysts in identifying short-term and long-term market trends using historical stock data.
4. **Risk Assessment:** Uses historical price movements and volatility-related features to understand market risk before making investment decisions.
5. **Financial Research:** Can be used by researchers and data analysts to study stock market behavior and evaluate machine learning models for price prediction.

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The dataset

The dataset was collected from **Yahoo Finance** using web scraping and contains historical stock market data for **three companies**. It includes the **Date, Company, Open, High, Low, Close, and Volume** columns. During feature engineering, I created additional features such as **Previous_Close, Close_Lag_3, Rolling_Mean_7, Rolling_STD_7, Closing_Price_Difference, and Daily_Return** to improve model performance. The **Close** column was used as the target variable, and the predicted closing price was compared with the previous closing price to determine whether the stock price was expected to move upward or downward.

Aim of the Project

The aim of this project is to predict whether the stock price is likely to move **upward or downward** using historical stock market data Historical price data is used to train machine learning models to predict the future **closing price**, and the predicted price is compared with the previous closing price to identify the expected market direction.